



Lab 02: Wireless Access Point

Access Point · Laptop0 · Laptop1 · Smartphone0 · Switch · PC1 · PC2

Lab #02

Topic: Wireless Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure a wireless Access Point with SSID 'Ahmed', WPA2-PSK and AES encryption.
- Connect Laptop0, Laptop1 and Smartphone0 wirelessly to the Access Point.
- Connect PC1 and PC2 via wired switch on the same network segment.
- Verify wireless clients can reach wired clients using ping.
- Relate to Network+ objective 2.4: Wireless technologies and configuration.

2. Network Topology

The topology has two sides: a wireless segment (Access Point with Laptop0, Laptop1 and Smartphone0 connected via dashed wireless links) and a wired segment (Switch connected to PC1 and PC2). The Access Point is wired to the Switch on Fa2/1.

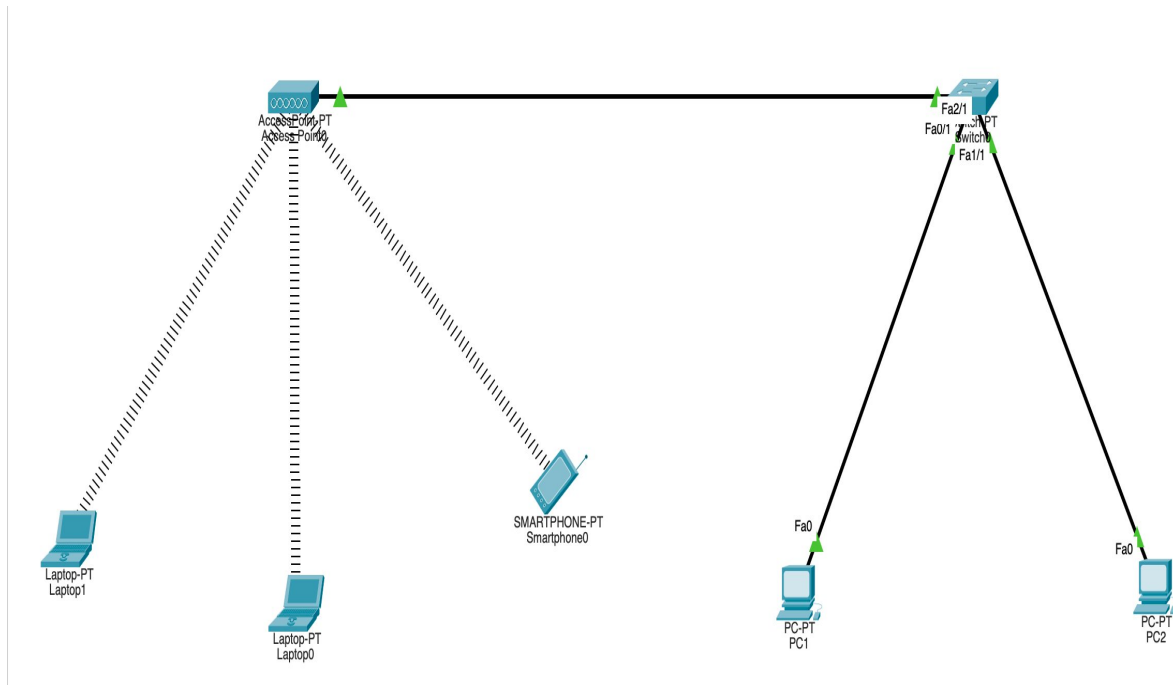


Figure 1 – Full topology: Access Point, wireless clients, Switch, PC1, PC2

Device	Type	Connection	SSID	IP Range
AccessPoint0	AccessPoint-PT	Wired to Switch Fa2/1	Ahmed	192.168.10.0/24
Laptop1	Laptop-PT	Wireless (WPA2)	Ahmed	DHCP



Laptop0	Laptop-PT	Wireless (WPA2)	Ahmed	DHCP
Smartphone0	SMARTPHONE-PT	Wireless (WPA2)	Ahmed	DHCP
PC1	PC-PT	Wired Fa0/1 Switch	—	192.168.10.x
PC2	PC-PT	Wired Fa1/1 Switch	—	192.168.10.x

3. Step-by-Step Configuration

3.1 Access Point – Port 1 Wireless Settings

Click AccessPoint0 → Config tab → Port 1 and enter the following settings as seen in the screenshot:

The screenshot shows the configuration window for Port 1 of an Access Point. The 'Config' tab is selected, and the 'Port 1' sub-tab is active. The configuration is as follows:

- Port Status: On (checked)
- SSID: Ahmed
- 2.4 GHz Channel: 6
- Coverage Range (meters): 140.00
- Authentication: WPA2-PSK (selected)
- Encryption Type: AES
- PSK Pass Phrase: 12345678

Figure 2 – Access Point Port 1 config: SSID Ahmed, WPA2-PSK, AES, PSK 12345678

Port	Port 1 (wireless)
SSID	Ahmed
2.4 GHz Channel	6
Coverage Range	140 metres
Authentication	WPA2-PSK
PSK Pass Phrase	12345678
Encryption Type	AES

3.2 Connecting Laptops – WPA2 Personal



On each laptop: Physical tab → install WPC300N wireless NIC. Then open PC Wireless → Connect tab → select SSID 'Ahmed' → enter passphrase:

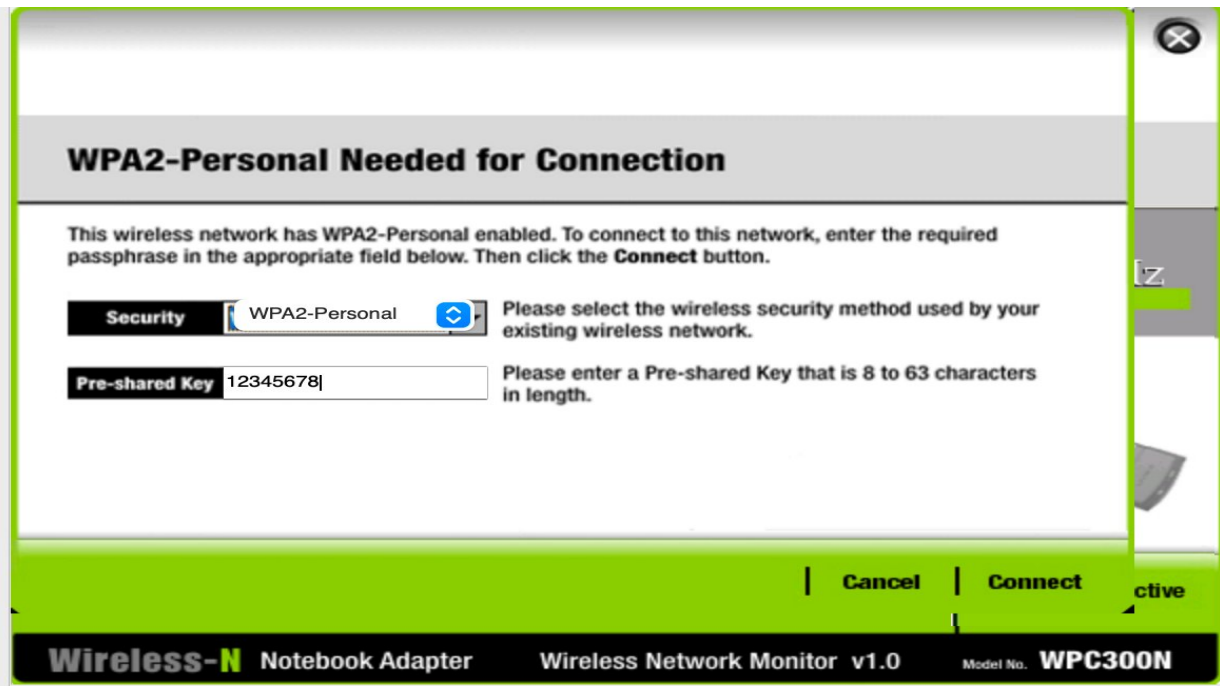


Figure 3 – WPA2-Personal connection prompt, Pre-shared Key: 12345678

1. Physical tab: remove Ethernet NIC, insert WPC300N wireless NIC
2. Desktop > PC Wireless > Connect tab
3. Click Refresh – SSID 'Ahmed' appears
4. Select 'Ahmed' > Connect
5. Security: WPA2-Personal Pre-shared Key: 12345678
6. Click Connect
7. Desktop > IP Configuration > set DHCP or static IP

4. Verification

```
Laptop0> ping 192.168.10.x      ! Ping PC1 or PC2 (wired)
Laptop1> ping 192.168.10.x      ! Confirm wireless-to-wired connectivity
Smartphone0> ping Laptop0       ! Wireless-to-wireless
```

5. Key Takeaways

- WPA2-PSK with AES is the current minimum security standard for Wi-Fi — never use WEP.
- The SSID 'Ahmed' identifies the wireless network; the PSK '12345678' authenticates clients.
- An Access Point bridges wireless to wired — it operates at Layer 2 like a switch.
- Channel 6 on 2.4 GHz is one of three non-overlapping channels (1, 6, 11).
- Coverage range (140 m) determines how far clients can be from the AP and still connect.